



Research Paper

Performance and methane emissions by beef heifer grazing in temperate pastures and in integrated crop-livestock systems: The effect of shade and nitrogen fertilization



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ABSTRACT

Methane (CH₄) emissions from enteric fermentation by cattle are the main source of greenhouse gases in livestock systems, but scarce information is available on the effect of production scenarios on CH₄ emissions. The objective of this study was to assess animal performance and CH₄ emissions by beef heifers grazing in a mixed *Lolium multiflorum* × *Avena strigosa* pasture in two different types of integrated systems, crop-livestock only (CL) or crop-livestock-tree (CLT), and with two nitrogen (N) supply levels (90 and 180 kg N ha⁻¹, N90 and N180, respectively). The study was performed during the winter season (*i.e.*, stocking season) over three years (2012–2014). Pasture shading by 6- to 8-year-old trees in the CLT system led to a reduction in the total annual herbage accumulation of approximately 2.6 Mg dry matter ha⁻¹ (–51%) and a reduction in winter carrying capacity of 0.5 stock units ha⁻¹ in comparison to the CL system. Average daily gain (ADG) was reduced by 32% in CLT compared to CL. Average CH₄ emissions were 163 ± 9.12 g CH₄ day⁻¹ in heifers with an initial average live weight (LW) of approximately 250 kg. Emissions of CH₄ per kg of LW did not differ between treatments (*P* > 0.05) with a mean of 0.58 ± 0.030 g kg⁻¹. However, there were significant differences between treatments and among years when CH₄ was expressed in g kg ADG⁻¹ and kg ha⁻¹ day⁻¹. Per unit area, CH₄ production ranged between 0.51 ± 0.05 (CLT N180) and 0.86 ± 0.12 kg ha⁻¹ day⁻¹ (CL N180). Possible strategies to reduce losses in animal production in the CLT systems are discussed, as well as the potential of C sequestration by woody biomass of eucalyptus trees to mitigate CH₄ emissions.

1. Introduction

Globally, interest in integrated crop-livestock systems as a strategy for sustainable land use has been increasing due to their potential to allow more income diversification and create fewer negative impacts on the environment (Lemaire et al., 2014; Moraes et al., 2014). The incorporation of a tree component in no-till, integrated crop-livestock systems may increase these systems' productivity and provide ecosystem services such as C sequestration both in the soil and in tree woody biomass (Udawatta and Jose, 2012). The growing market for animal products that have been produced under humane conditions has further increased the interest in this kind of system (Yates et al., 2007).

Therefore, integrated crop-livestock systems with trees present opportunities for important contributions to global food security and sustainable livelihoods.

Tree cover tends to ameliorate extremes in temperature and reduce wind speed, which is a benefit to animal welfare (Yates et al., 2007; Lopes et al., 2016). However, changes in the microclimate become more pronounced as the trees grow, which might also affect the growth of the understory crop (Lin et al., 1999). Tree density, spacing and management (*e.g.*, pruning and thinning) are other factors that determine how rapidly these changes come into play (Lin et al., 1999; Peng et al., 2009). Therefore, the issue is to quantify these changes in understory crop composition and microclimate caused by tree cover and the

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resulting effects on animal production and provision of ecosystem services.

The relative effects of shading on the production and nutritive value of grasses are not well described because understory plants can exhibit alterations in anatomy and physiology to compensate for low light quantity and distinct quality (Niinemets et al., 1998; Cavagnaro and Trione, 2007) and consequently increase or decrease herbage yield and nutritive value (Lin et al., 2001; Pontes et al., 2016). For example, some studies have suggested negative effects of shading on nutritive value (Wilson and Wong, 1982; Samarakoon et al., 1990) due to a decrease in the soluble carbohydrate level in plants accompanied by an increase in cell wall content, which reduces overall digestibility. In contrast to these findings, there are reports of higher nutritive value – primarily from increases in crude protein content – in plants grown in shade than in plants grown in full sunlight conditions (Lin et al., 2001; Peri et al., 2007; Abraham et al., 2014). It is worth noting that species responses to shading are also dependent on nutrient availability, particularly nitrogen (Pontes et al., 2016). These inconsistent responses lead to differences in the quantity and quality of the ingested herbage and thus alter animal performance and methane (CH₄) emissions.

Understanding how integrated crop-livestock systems may contribute to the reduction of greenhouse gases is essential for the development of public policies for environmental preservation (Savian et al., 2014). The need to mitigate ruminant CH₄ emissions is particularly important in Brazil, which is the second largest beef producer in the world (Paulino and Duarte, 2014). However, CH₄ emissions in integrated crop-livestock systems with trees have yet to be comprehensively studied (Molina et al., 2016).

According to Shibata and Terada (2010), the most promising and cost-effective way to reduce CH₄ emissions from cattle is to improve the efficiency of livestock production by, for example, improving feed quality and livestock management. For instance, grazing on less-mature pastures with lower fiber and higher soluble carbohydrates reduces CH₄ production (see Eckard et al., 2010 for a review). This tactic may help reduce CH₄ emissions per unit of animal product (Knap et al., 2014). Regardless, multiple strategies to mitigate greenhouse gas emissions should be evaluated on a broad scale while also considering the context of each production system (Berndt and Tomkins, 2013).

This study focuses on animal performance and CH₄ emissions from cattle grazing on mixed Italian ryegrass (*Lolium multiflorum* L.) × black oat (*Avena strigosa* Schreb.) pasture (i.e., two C₃ cool-season grasses) in rotation with soybean (*Glycine max* L. Merrill) or maize (*Zea mays* L.) in two integrated crop-livestock systems, with and without trees and at two N supply levels (90 and 180 kg N ha⁻¹). This study was performed over three years (2012–2014) in 6–8-year-old tree plantations growing in a subtropical climate. We hypothesized that the responses of understory species to combined shade and N effects provoke differences in the quantity and quality of the herbage ingested and thus alter animal production and CH₄ emissions from grazing animals. Our results provide data on animal production and CH₄ emissions to assist in the evaluation of tree inclusion in integrated crop-livestock systems. We also discuss the potential of C sequestration by woody biomass to further mitigate CH₄ emissions from cattle.

2. Materials and methods

2.1. Study area

A field experiment was conducted at the Agronomic Institute of Paraná, Ponta Grossa-PR (25°07'22"S, 50°03'01"W) in Southern Brazil. The local climate is humid subtropical, or Cfb in the Köppen classification system (Köppen and Geinger, 1928), with frequent occurrence of frosts and a mean annual temperature of 17.6 °C ranging from 14 °C in July to 21 °C in January. The mean annual rainfall is 1400 mm. The soil is a Cambisol and Ferralsol (Embrapa, 2006) with 19%, 3% and 78% clay, silt and sand in the upper 20 cm, respectively. In 2006, three tree

species (eucalyptus, *Eucalyptus dunnii*; pink pepper, *Schinus molle*; and silver oak, *Grevillea robusta*) were planted in 6 of 12 experimental units. The species were interspersed in the same rows running crosswise in relation to the slope with a predominantly east-west orientation at 3 × 14 m spacing (238 trees ha⁻¹). The experimental area had been managed as an integrated crop-livestock system using no-till since 2009. Since winter 2010, the production system integrated cattle grazing on cool-season pasture (black oat + ryegrass) and warm-season maize (cv. IPR164) or soybean (cv. BRS232) crops on the same area and in the same cropping year. During the summer of 2013, the experimental area was thinned to 159 trees ha⁻¹ by removing pink pepper trees, many of which had been damaged by cattle activity (Porffrio-da-Silva et al., 2012).

The black oat + ryegrass mixture was sown each year in rows with a seeding density of 45 and 15 kg ha⁻¹, respectively, in May/2012, April/2013 and May/2014 and 400 kg ha⁻¹ of commercial formula 4-30-10 (N-P-K) was applied. Fertilizers were also applied annually for maize (400 kg ha⁻¹ of commercial formula 10-30-10 + 270 kg urea ha⁻¹) or soy (400 kg ha⁻¹ of commercial formula 0-20-20). Soybean and maize were sown in alternate years at the beginning of November.

2.2. Treatments and the experimental design

The experimental design was a randomized complete block with three replicates (paddocks). Two nitrogen fertilization treatments (90 and 180 kg N ha⁻¹, or N90 and N180, respectively) were crossed with two integrated crop-livestock systems: crop-livestock only (CL) and crop-livestock with trees (CLT). The four resulting treatments were: CL N90, CL N180, CLT N90 and CLT N180. Nitrogen fertilizer (as urea) was applied in a single application during every stocking season at ~40 days after pasture was sown. The fertilizer N supply was chosen to provide limiting (N90) and non-limiting (N180) N nutrition according to Alves (2002).

The experimental area of 13.07 ha was divided into 12 paddocks ranging from 0.77 to 1.22 ha (see Fig. 1). Each experimental unit (paddock) received three tester animals (permanent animals that remained throughout the experimental period) and a variable number of animals periodically adjusted to maintain the desired sward height of 20 cm (“put-and-take” method, Mott and Lucas, 1952). According to Kunrath et al. (2014), managing mixed annual ryegrass and black oat pastures at 20 cm under continuous stocking leads to several soil-plant-animal system benefits. This management criterion was adopted for both CL and CLT to allow for inter-comparability although no information is available regarding optimal sward height for ruminant weight gain and forage production for the species studied in association with trees.

The experimental animals were Purunã (¼ Aberdeen Angus, ¼ Canchim, ¼ Caracu, and ¼ Charolais) beef heifers with an average age

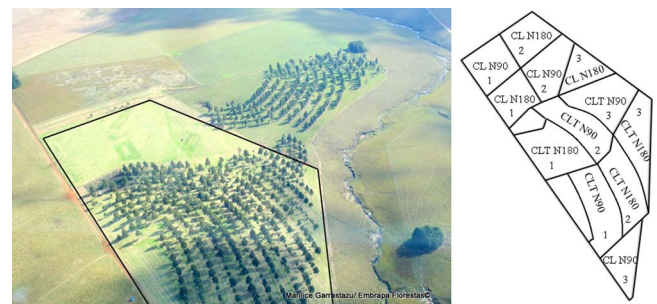


Fig. 1. The aerial view of the experimental site at Ponta Grossa, Paraná, Brazil (Photo: Marilice Garrastazu, Embrapa Florestas). The marked area in the picture is the map of the experimental area on the right with the 12 paddocks and their treatments: two integrated crop-livestock systems, i.e., crop-livestock only (CL) and crop-livestock with trees (CLT), and two N levels, 90 and 180 kg N ha⁻¹, i.e., N90 and N180, respectively. The numbers indicate the three blocks.

of 10 months and weighing 269 ± 5.7 , 256 ± 5.1 and 224 ± 6.8 kg for 2012, 2013 and 2014, respectively. Stocking season began on July 30th in 2012, July 3rd in 2013 and July 18th in 2014 with grazing starting at approximately two months after pasture sowing, i.e., when all experimental units reached an average of at least 25 cm height for 2012 and 2013 and 20 cm for 2014. Heifers grazed until September 12th in 2012, October 8th in 2013 and October 17th in 2014 and totaled 44, 97 and 91 grazing days for 2012, 2013 and 2014, respectively. The short stocking season in 2012 was caused by a severe drought in August (rainfall of only 5 mm) that undermined the pasture productivity. All animals had unrestricted access to shade in CLT plots, and clean water and mineral supplements throughout the experiment in CLT and CL plots.

2.3. Tree measurements and shading percentage

Tree heights and diameters at breast height (1.3 m) were measured in April 2011 and near the end of the experiment in October 2014. In 2011, 7 trees per species and within each paddock were measured ($n = 168$). In 2014, 10% of the eucalyptus trees ($n = 30$) and all of the silver oak trees were evaluated ($n = 305$). Shading was measured using two ceptometers (Decagon LP-80 AccuPAR, Pullman, WA, USA) with one placed in full sun and the other placed between the rows of trees (1.4, 4.2, 7, 9.8 and 12.6 m from the row). Shading percentage measurements were taken every 30 min from 9:00 to 15:00 h at 1 m above ground level. The decrease in light availability for the understory vegetation was calculated as the difference in the values from these two ceptometers.

2.4. Pasture measurements

Sward height (SH) was measured at 100 points per paddock every 15 days using a sward stick (Barthram, 1985). These measurements were made at random across grazed and ungrazed areas, only avoiding areas around gates, watering points and resting sites as in Hodgson (1990).

Herbage mass (HM, kg ha⁻¹ of dry matter, DM), daily herbage accumulation rate (HAR, kg DM ha⁻¹ day⁻¹) and total herbage accumulation (THA, kg DM ha⁻¹ year⁻¹) were estimated from samples collected every ~21 days, totaling three cutting dates for 2012 and five for 2013 and 2014. The herbage mass was estimated in each paddock from five cuts above the litter mass within a quadrat measuring 0.25 m² (50 × 50 cm). The cuts were made randomly inside the total area of each paddock. From these samples, sub-samples were manually separated into fractions containing leaf blade, stem + sheath, dead material and other species. Leaf proportion was determined as the leaf blade mass in relation to the weight of the sub-sample. Samples and sub-samples were dried at 60 °C for 48 h and then weighed. Herbage accumulation rates were determined using three grazing enclosure cages in each paddock (Klingman et al., 1943). The herbage allowance (HA, % LW) was calculated according to the following equation: $HA (\% LW) = ((HM/n + HAR)/SR) \times 100$, where LW = live weight, HM = average herbage mass of each stocking period (kg DM ha⁻¹), n = number of days of the stocking period and SR = stocking rate of each stocking period (kg LW ha⁻¹) (Sollenberger et al., 2005).

Total herbage accumulation was calculated as the sum of the accumulated herbage during each period (accumulation rate × number of days in period) and the herbage mass at the beginning of the grazing phase (Kunrath et al., 2014).

Samples harvested on August 18, 2012, and August 8 and September 3, 2013, for the HM measurements were also milled through a 1-mm screen and analyzed for crude protein (CP), dry-matter digestibility (DMD), neutral detergent fiber (NDF) and acid detergent fiber (ADF) via near-infrared reflectance spectroscopy (NIRS) (FOSS-NIRSystems 5000, scanning over the spectral range of 1100–2500 nm; CEPA laboratory, Passo Fundo-RS, Brazil). The calibration models

obtained for the herbage CP, NDF, ADF and DMD concentrations included the following statistical parameters: number of samples (444, 271, 393 and 369); mean (9.9, 64.1, 34.8 and 93.0); standard error of cross-validation (0.6, 1.3, 1.3 and 0.7); and cross-validation R^2 (0.98, 0.97, 0.97 and 0.94).

2.5. Animal performance

Animals were weighed every ~21 days (totaling two stocking periods for 2012 and four for 2013 and 2014) after fasting from solids for approximately 15 h. The stocking rate (SR), expressed in kg ha⁻¹, was calculated by adding the average live weight of the testers with the average live weight of each put-and-take animal, multiplying by the number of days they remained in each paddock and dividing by the total number of grazing days in each stocking period. The average daily gain (ADG, g animal⁻¹ day⁻¹) was calculated as the difference between the final and initial weights of the tester animals divided by the number of days in each stocking period. The gain per hectare (Gha, kg ha⁻¹) was obtained by multiplying the animals ha⁻¹ by the ADG of the tester animals and the number of total grazing days.

2.6. Estimates of daily methane emission

Daily CH₄ emission was measured using the SF₆ tracer technique described by Johnson et al. (1994) with adaptations (Gere and Gratton, 2010). The SF₆ permeation tubes used in the experiment had an average permeation rate of 2.980 ± 0.6059 mg day⁻¹. Methodological details are described in Savian et al. (2014). Gas collections were performed each year on two animals per paddock (a total of 24 Purunã beef heifers per year) over 5 consecutive days starting on August 9, 2012, August 15, 2013, and August 27, 2014. Methane emissions per kg of live weight, per kg of ADG and per area were determined using daily CH₄ emissions per animal, ADG, and SR, respectively.

2.7. Statistical analyses

Analyses of variance were performed for animal- and pasture-related variables using a split-split-plot-like or split-plot-like model in Statgraphics Centurion XV (2006). In the split-plot-like model, main plots were the treatments (nitrogen fertilization*crop-livestock system, DF = 3) and subplots were years (DF = 2), whereas in the split-split-plot-like model, main plots were treatments, subplots were years (DF = 1) and sub-subplots were stocking periods (DF = 3, for HAR, HA, SH, ADG and SR) or cutting dates (DF = 4, for leaf proportion and HM). All factors except blocks were considered as fixed terms (Pons et al., 2005). The error term for treatments effect was block(treatment) and the subplot error was year*block(treatment). Interactions were checked for each variable and removed from the model if they had a P -value > 0.05. When necessary, the data were transformed *via* log, box-cox or arcsine of the square root to normalize the residuals. Daily CH₄ emissions data were subjected to an outlier analysis (Grubbs' Extreme Studentized Deviate Test) and points were eliminated when $P < 0.05$. An additional five animals were excluded from the analysis for exhibiting unusually low daily CH₄ emissions (44 ± 5.80 g CH₄ animal⁻¹ day⁻¹, on average), considered beyond the physiological limits described in the literature.

Two sets of data were analyzed using means per paddock: i) means from 2013 and 2014 to characterize animal production, eliminating the short 2012 stocking season to provide a reliable estimate; and ii) data from periods close to the dates for gas collection to correlate animal and pasture variables with CH₄ emissions. In other words, in 2012, we considered animal-related variables obtained in the first stocking period (i.e., between July 30th and August 21st), whereas for 2013 and 2014 we considered data obtained in the third (i.e., between August 15th and September 6th) and second (i.e., between August 7th and September 2nd) stocking periods, respectively. Pasture-related variables were

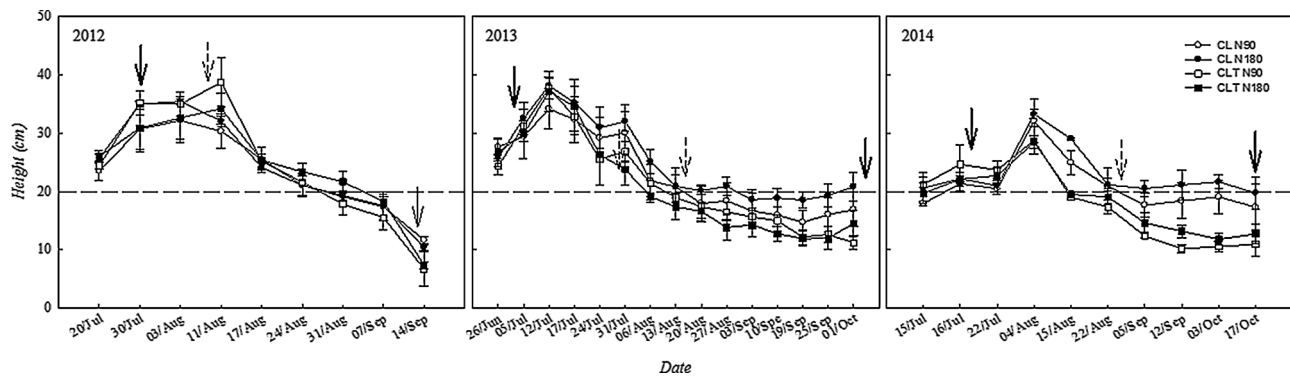


Fig. 2. The sward height means (cm, $n = 3$) over time for the three experimental years (2012, 2013 and 2014). The dashed line indicates the target sward height (20 cm). Solid arrows indicate the start and the end of the grazing period. Dashed arrows indicate the start of gas collection. CL, integrated crop-livestock system; CLT, integrated crop-livestock system with trees; N90, 90 kg N ha⁻¹ and N180, 180 kg N ha⁻¹.

estimated by considering the samples collected close to these periods.

3. Results

3.1. Tree measurements and shading percentages

In April 2011, the average ($\pm$ standard error) tree height was 12.8 ± 0.3 m and 8.1 ± 0.2 m and the average diameters at breast height were 22.5 ± 0.9 cm and 11.5 ± 0.4 cm for eucalyptus and silver oak, respectively (Porfírio-da-Silva, 2012). By October 2014, the trees had reached a height of 22.1 ± 0.36 m and 12.3 ± 0.14 m and diameters at breast height of 43.0 ± 0.85 cm and 20.4 ± 0.24 cm for eucalyptus and silver oak, respectively. During the experimental years, the mean percentage of light reduction under the tree canopy compared to the treeless system was $39 \pm 1.45\%$ in 2012, $41 \pm 1.10\%$ in 2013 and $53 \pm 1.49\%$ in 2014.

3.2. Sward height

Fig. 2 shows the variations of SH over time within each treatment. The stocking period started when pastures reached an average of 33 ± 1.4 , 31 ± 1.2 and 22 ± 0.7 cm in 2012, 2013 and 2014, respectively. In September of 2012, a large reduction in SH for all treatments was observed (Fig. 2), which led to a shorter experimental period. At the end of the stocking period, the SH for the CL systems was close to the target SH both in 2013 and 2014. It was more difficult to maintain the target SH in the CLT systems where SH remained below 20 cm, particularly after September, and reached ~ 13 cm by the end of the stocking period.

3.3. Sward characteristics and animal production

An analysis of variance for pasture and animal data showed significantly greater values in the CL compared to the CLT systems regardless of N level for all variables except SH and LP (Table 1). Differences among years (DF = 1) were observed for HM, HA, THA, Gha and LP (Table 2). Despite significantly greater values for HM, HA and THA in 2013, the Gha and the LP were greater in 2014 (Table 2). The interaction between treatments and year (DF = 3) was significant for HM, HA and HAR, with greater differences among years and lower values in 2014 for CLT systems (Table A.1 in the Supplementary material).

Significant ($P < 0.001$) differences among stocking periods (DF = 3 or 4) were observed for all variables (Table 3). This factor was the most important, in terms of variance explained, for SH (57%), HAR (36%), SR (39%) and LP (85%). In general, we observed a reduction in all these values through the experimental period (Table 3). Significant interactions between the treatment and stocking period/cutting date

were observed for SH, HA (DF = 9) and HM (DF = 12), and between year and stocking period/cutting date for SH, HAR, ADG, SR (DF = 3) and HM (DF = 4). Means for these interactions are available in Tables A.2 and A.3, respectively (in the Supplementary material). For SH and HA, reduction in values throughout the experimental period were most important in CLT systems, particularly in the last two cutting dates for HM (Table A.2 in the Supplementary material).

3.4. Sward characteristics and animal production during gas collection periods

Differences between treatments during gas collection periods were significant for HM, SR, ADG and Gha (Table 4). For HM and SR, we observed greater values in CL compared to CLT systems regardless of N level (Table 4). Differences between treatments for ADG reached ~ 414 g animal⁻¹ day⁻¹ (Table 4). Although Gha increased with N level in the CL system, the same was not observed in the CLT system (Table 4).

The year (DF = 2) effect was significant for all variables (Table 5). In general, we observed greater values in the period evaluated in 2012 (except for ADG, Table 5). The interaction between year and the treatment (DF = 6) was significant ($P < 0.05$) for SH, HM, HAR and SR as shown in Table A.4 (in the Supplementary material). Differences between treatments for SH were more important in 2014 compared to 2012 and 2013 (Table A.4 in the Supplementary material).

No significant differences were observed between treatments for nutritive value parameters (Table 4). Significant differences among cutting dates (DF = 2) were observed only for ADF ($F = 7.5$, $P < 0.01$), with greater values observed in the first cutting date ($38 \pm 1.0\%$) compared to the second ($34 \pm 0.5\%$) and third cutting date ($35 \pm 1.1\%$). No significant interactions were observed among the factors studied for the nutritive value parameters ($P > 0.05$).

3.5. Methane emissions

Emissions of CH₄ per animal and per kg of live weight did not differ among treatments and years ($P > 0.05$, Tables 4 and 5), with a mean of 163 ± 9.12 g animal⁻¹ day⁻¹ and 0.58 ± 0.030 g kg⁻¹ LW ($n = 42$), respectively. CH₄ emitted per unit area was significantly greater in the CL N180 treatment ($P < 0.10$, Table 4). CH₄ emissions per ADG and CH₄ emitted per unit area were significantly different among years (DF = 2, $P < 0.10$), with lower values observed in 2014 and 2013, respectively (Table 5).

Table 1

Pasture and animal performance variables (mean $\pm$ standard error) for beef heifers grazing on mixed *Lolium multiflorum* $\times$ *Avena strigosa* under different integration and nitrogen treatments (CL, crop-livestock and CLT, crop-livestock-trees systems; 90 and 180 kg N ha⁻¹, N90 and N180, respectively).

	CL N90	CL N180	CLT N90	CLT N180	F	P
SH (cm)	22 $\pm$ 1.3ab	24 $\pm$ 1.0a	18 $\pm$ 1.4b	19 $\pm$ 1.4b	3.1	0.0924
HM (kg DM ha ⁻¹)	1816 $\pm$ 141.8a	2214 $\pm$ 141.3a	854 $\pm$ 68.9b	819 $\pm$ 73.9b	17	< 0.001
HAR (kg DM ha ⁻¹ day ⁻¹)	46 $\pm$ 5.8a	47 $\pm$ 7.0a	23 $\pm$ 4.0b	19 $\pm$ 2.8b	8.6	0.0071
HA (%)	11 $\pm$ 1.0a	12 $\pm$ 0.8a	6.3 $\pm$ 0.7b	5.8 $\pm$ 0.7b	6.7	0.0144
THA (kg DM ha ⁻¹ year ⁻¹)	5087 $\pm$ 628.9a	5299 $\pm$ 262.0a	2681 $\pm$ 375.8b	2373 $\pm$ 257.5b	9.6	0.0050
SR (kg LW ha ⁻¹)	1032 $\pm$ 43.7a	1056 $\pm$ 43.5a	824 $\pm$ 42.0b	839 $\pm$ 42.4b	4.6	0.0374
ADG (g animal ⁻¹ day ⁻¹)	882 $\pm$ 86.9a	937 $\pm$ 67.5a	567 $\pm$ 72.5b	664 $\pm$ 87.4b	9.4	0.0053
Gha (kg LW ha ⁻¹ day ⁻¹)	3.4 $\pm$ 0.30a	3.9 $\pm$ 0.30a	1.8 $\pm$ 0.22b	2.1 $\pm$ 0.16b	26	< 0.001
LP (%)	43 $\pm$ 4.3a	42 $\pm$ 4.4a	41 $\pm$ 4.6a	44 $\pm$ 4.3a	0.9	0.4466

F and P = F-ratio and probability, respectively, for the treatment effect; SH = sward height (n = 24); HM = herbage mass (n = 30); HAR = herbage accumulation rate (n = 24); HA = herbage allowance (n = 24); THA = total herbage accumulation (n = 6); SR = stocking rate (n = 24); ADG = average daily gain (n = 24); Gha = gain per hectare (n = 6); LP = leaf proportion (n = 27); DM = dry matter; LW = live weight. Within rows, means followed by the same letter are not significantly different according to LSD test (P < 0.05).

Table 2

Pasture and animal performance variables (mean $\pm$ standard error) for beef heifers grazing on mixed *Lolium multiflorum* $\times$ *Avena strigosa* by year.

	2013	2014	F	P
SH (cm)	22 $\pm$ 1.1	19 $\pm$ 0.8	0.1	0.7582
HM (kg DM ha ⁻¹)	1649 $\pm$ 114.7	1202 $\pm$ 99.8	32	< 0.001
HAR (kg DM ha ⁻¹ day ⁻¹)	35 $\pm$ 4.4	33 $\pm$ 3.7	0	0.9723
HA (%)	11 $\pm$ 0.7	7.0 $\pm$ 0.6	41	< 0.001
THA (kg DM ha ⁻¹ year ⁻¹)	4170 $\pm$ 422.6	3550 $\pm$ 531.7	8.5	0.0139
SR (kg LW ha ⁻¹)	937 $\pm$ 26.7	939 $\pm$ 39.7	0	0.9591
ADG (g animal ⁻¹ day ⁻¹)	722 $\pm$ 59.1	803 $\pm$ 59.5	0.3	0.5990
Gha (kg LW ha ⁻¹ day ⁻¹)	2.5 $\pm$ 0.21	3.1 $\pm$ 0.38	6.4	0.0285
LP (%)	38 $\pm$ 2.6	49 $\pm$ 3.5	48	< 0.001

F and P = F-ratio and probability, respectively, for the year effect; SH = sward height (n = 48); HM = herbage mass (n = 30); HAR = herbage accumulation rate (n = 24); HA = herbage allowance (n = 24); THA = total herbage accumulation (n = 6); SR = stocking rate (n = 24); ADG = average daily gain (n = 24); Gha = gain per hectare (n = 6); LP = leaf proportion (n = 27); DM = dry matter; LW = live weight.

4. Discussion

4.1. Animal production

Generally, livestock production in the CLT systems is equal to that for open pastures during the first years of tree growth (e.g., Teklehaimanot et al., 2002). In fact, previous results obtained in the same experimental area during the winters of 2010 and 2011 did not show differences between the systems for ADG and Gha (Porfírio-da-Silva, 2012). However, once the tree canopy closes, forage production and thus animal production may decline (Huang and Xu, 1999; Silva-Pando et al., 2002). The challenge is to quantify these yield losses and pinpoint when they begin to occur. For instance, the 6- to 8-year-old trees in our experiment reduced beef heifer gains (ADG) on black oat

+ ryegrass mixture by 32% compared to pastures without trees. Yates et al. (2007) and Lopes et al. (2016) suggested that trees in the CLT systems protect animals from wind and extreme temperatures, which in turn can mitigate the effects of lower forage production. However, the competitive effect of trees on pasture growth was so strong in our study that any positive effect of trees on environmental conditions was not reflected in animal performance.

Lin et al. (1999) stated that some forage species, mainly C₃ perennial grasses grown at 50% sunlight, can sustain productivity at rates comparable to those obtained in full-sun conditions. However, in the present study, important differences were observed in the HM and consequently in the HAR and THA for the two systems during the experimental period, even with increasing N availability. This result indicates that shade provided by trees in the CLT – as high as 39% in relation to the open field – affected pasture growth.

Perennial grasses may have a better capacity to adapt their phenotype to cope with shaded environments compared to annual grasses. The annual grasses studied here allocated carbon towards stem elongation at the expense of leaf development and tillering (see Pontes et al., 2017), which reduced the herbage production similarly to other authors' observations (Kirchner et al., 2010; Belesky et al., 2011). On average, the HM in the CLT system remained below the value considered to be the minimum (1200 kg ha⁻¹ according to Hodgson, 1990) to ensure adequate intake for grazing animals, and animal performance suffered as a consequence. This result is particularly important for no-till integrated crop-livestock systems due to the dual purpose of HM, i.e., maximizing animal production and generating enough soil cover for the succeeding crop. Therefore, the HM in the CLT system (~837 kg ha⁻¹) is too low to maintain the long-term performance of no-till operations (Kunrath et al., 2014).

Lower animal performance in the CLT systems could be associated with both the shade effect and difficulty in maintaining the target SH (Fig. 2). However, even in periods when the SH in the CLT systems was

Table 3

Pasture and animal performance variables (mean $\pm$ standard error) for beef heifers grazing on mixed *Lolium multiflorum* $\times$ *Avena strigosa* by stocking period.

	1st	2nd	3rd	4th	5th	F	P
SH (cm)	28 $\pm$ 1.0a	22 $\pm$ 0.8b	17 $\pm$ 0.8c	15 $\pm$ 0.9c	–	142	< 0.001
HM (kg DM ha ⁻¹)	883 $\pm$ 57.6b	1642 $\pm$ 144.3a	1699 $\pm$ 187.6a	1423 $\pm$ 200.8a	1480 $\pm$ 202.8a	37	< 0.001
HAR (kg DM ha ⁻¹ day ⁻¹)	50 $\pm$ 6.1a	38 $\pm$ 4.1b	26 $\pm$ 4.4c	14 $\pm$ 4.6d	–	35	< 0.001
HA (%)	10 $\pm$ 1.0a	9.1 $\pm$ 0.7a	9.9 $\pm$ 1.1a	6.5 $\pm$ 0.8b	–	20	< 0.001
SR (kg LW ha ⁻¹)	1020 $\pm$ 35.0b	1134 $\pm$ 47.3a	805 $\pm$ 32.3c	793 $\pm$ 34.5c	–	50	< 0.001
ADG (g animal ⁻¹ day ⁻¹)	756 $\pm$ 81.1ab	642 $\pm$ 84.7b	956 $\pm$ 97.3a	697 $\pm$ 57.9b	–	6.2	< 0.001
LP (%)	77 $\pm$ 0.9a	51 $\pm$ 2.2b	23 $\pm$ 1.3d	29 $\pm$ 1.7c	23 $\pm$ 2.1d	240	< 0.001

F and P = F-ratio and probability, respectively, for the stocking period effect; SH = sward height; HM = herbage mass; HAR = herbage accumulation rate; HA = herbage allowance; SR = stocking rate; ADG = average daily gain; Within rows, means (n = 24) followed by the same letter are not significantly different according to LSD test (P < 0.05). 2013: 1st = July 3rd–25th; 2nd = July 25th–August 15th; 3rd = August 15th–September 6th; 4th = September 6th–October 8th. 2014: 1st = July 18th–August 7th; 2nd = August 7th–September 2nd; 3rd = September 2nd–23th; 4th = September 23th–October 17th. In 2013, samples for HM and LP were cut on the 28th of June, 17th of July, 8th of August and 3rd and 30th of September. In 2014, samples were cut on the 16th of July, 4th and 25th of August and 19th of September.

Table 4

Pasture and animal performance variables (mean $\pm$ standard error) measured during gas collection periods and methane (CH₄) emissions for beef heifers grazing on mixed *Lolium multiflorum* $\times$ *Avena strigosa* under different integration and nitrogen treatments (CL, crop-livestock and CLT, crop-livestock-trees systems; 90 and 180 kg N ha⁻¹, N90 and N180, respectively).

	CL N90	CL N180	CLT N90	CLT N180	F	P
SH (cm)	23 $\pm$ 2.2a	25 $\pm$ 1.6a	23 $\pm$ 2.7a	22 $\pm$ 2.5a	0.96	0.4582
HM (kg DM ha ⁻¹)	1970 $\pm$ 221.2a	2351 $\pm$ 190.3a	1247 $\pm$ 149.0b	1124 $\pm$ 91.8b	10.8	0.0034
HAR (kg DM ha ⁻¹ day ⁻¹)	65 $\pm$ 10.6a	62 $\pm$ 7.9a	34 $\pm$ 4.3a	30 $\pm$ 4.8a	2.7	0.1185
HA (%)	12 $\pm$ 1.7a	13 $\pm$ 1.4a	10 $\pm$ 0.8a	9.0 $\pm$ 0.9a	0.94	0.4636
SR (kg LW ha ⁻¹)	1366 $\pm$ 82.7a	1509 $\pm$ 162.0a	940 $\pm$ 74.6b	956 $\pm$ 90.7b	20.4	< 0.001
ADG (g animal ⁻¹ day ⁻¹)	872 $\pm$ 92.1ab	1053 $\pm$ 37.8a	639 $\pm$ 71.9c	693 $\pm$ 62.2bc	8.3	0.0078
Gha (kg LW ha ⁻¹ day ⁻¹)	4.5 $\pm$ 0.66b	5.7 $\pm$ 0.67a	2.6 $\pm$ 0.40c	2.7 $\pm$ 0.47c	18.3	< 0.001
LP (%)	46 $\pm$ 5.8a	43 $\pm$ 6.5a	42 $\pm$ 5.8a	45 $\pm$ 6.0a	0.98	0.4475
CP (%)	15 $\pm$ 0.5a	16 $\pm$ 0.3a	14 $\pm$ 0.4a	15 $\pm$ 0.5a	1.21	0.3670
DMD (%)	64 $\pm$ 1.3a	63 $\pm$ 1.4a	61 $\pm$ 1.5a	61 $\pm$ 1.1a	1.46	0.2103
NDF (%)	63 $\pm$ 0.9a	65 $\pm$ 0.6a	65 $\pm$ 0.5a	64 $\pm$ 0.9a	0.71	0.5729
ADF (%)	34 $\pm$ 0.8b	36 $\pm$ 0.6ab	38 $\pm$ 1.3a	36 $\pm$ 1.3ab	2.76	0.1118
g CH ₄ animal ⁻¹ day ⁻¹	170 $\pm$ 13.2a	166 $\pm$ 25.0a	165 $\pm$ 17.7a	155 $\pm$ 18.1a	0.20	0.8935
g CH ₄ kg LW ⁻¹	0.60 $\pm$ 0.06a	0.57 $\pm$ 0.07a	0.59 $\pm$ 0.05a	0.57 $\pm$ 0.07a	0.24	0.8655
g CH ₄ kg ADG ⁻¹	247 $\pm$ 32.9ab	168 $\pm$ 27.2b	286 $\pm$ 36.4a	236 $\pm$ 35.1ab	1.99	0.1945
kg CH ₄ ha ⁻¹ day ⁻¹	0.78 $\pm$ 0.08a	0.83 $\pm$ 0.12a	0.54 $\pm$ 0.07b	0.51 $\pm$ 0.05b	3.99	0.0521

F and P = F-ratio and probability, respectively, for the treatment effect; SH = sward height; HM = herbage mass; HAR = herbage accumulation rate; HA = herbage allowance; SR = stocking rate; ADG = average daily gain; Gha = gain per hectare; LW = live weight; LP = leaf proportion; CP = crude protein; DMD = dry-matter digestibility; NDF = neutral detergent fiber; ADF = acid detergent fiber; Within rows, means (n = 9) followed by the same letter are not significantly different according to LSD test (P < 0.05).

similar to the CL system, a reduction of ~27% in ADG and ~58% in Gha was observed (see N90 treatments, Table 4). In fact, despite small differences in SH between treatments, considerable differences were observed in HM (Table 1). Belesky et al. (2011) argued that plants growing in shaded sites quickly elongate leaves, which allows plants to attain the target SH but with a weakened structure and fewer tillers compared to plants growing in full sun systems. Because SR was controlled by SH, SR was satisfactory during periods when the target SH was attained in the CLT system and according to values reported by other authors for black oat + ryegrass mixtures in full sunlight conditions and with the same target SH (e.g., ~792 kg LW ha⁻¹, Carvalho et al., 2010). However, the HA between the CL and the CLT systems was different even with similar SH.

The large increase in shading percentage in 2014, despite thinning in the previous year, intensified the negative effect of the presence of trees on sward characteristics. For instance, daily HA was drastically reduced in 2014 (−4.0%, Table 2). An HA lower than 3–4 times the potential of animal consumption limits intake (Sollenberger et al., 2005), which suggests that silvicultural interventions need to be intensified to minimize tree-crop competition and thus facilitate the maintenance of the target SH, ensure high animal performance, and maintain adequate HM residues for soil cover for the subsequent crop.

Table 5

Pasture and animal performance variables (mean $\pm$ standard error) measured during gas collection periods and methane (CH₄) emissions for beef heifers grazing on mixed *Lolium multiflorum* $\times$ *Avena strigosa* by year.

	2012	2013	2014	F	P
SH (cm)	31 $\pm$ 0.34a	17 $\pm$ 1.0c	21 $\pm$ 1.0b	128	< 0.001
HM (kg DM ha ⁻¹)	1761 $\pm$ 112.8a	1948 $\pm$ 273.9a	1309 $\pm$ 161.1b	14	< 0.001
HAR (kg DM ha ⁻¹ day ⁻¹)	56 $\pm$ 8.2a	35 $\pm$ 6.5b	51 $\pm$ 7.4a	14	< 0.001
HA (%)	11 $\pm$ 0.9a	13 $\pm$ 1.4a	9.0 $\pm$ 0.65b	4.2	0.0015
SR (kg LW ha ⁻¹)	1444 $\pm$ 101.4a	868 $\pm$ 60.9c	1265 $\pm$ 113.3b	38	< 0.001
ADG (g animal ⁻¹ day ⁻¹)	777 $\pm$ 52.9b	711 $\pm$ 81.7b	955 $\pm$ 71.8a	6.3	0.0069
Gha (kg LW ha ⁻¹ day ⁻¹)	4.5 $\pm$ 0.44a	2.1 $\pm$ 0.31b	5.0 $\pm$ 0.64a	27	< 0.001
LP (%)	66 $\pm$ 0.70a	25 $\pm$ 1.64c	41 $\pm$ 1.12b	265	< 0.001
g CH ₄ animal ⁻¹ day ⁻¹	158 $\pm$ 11.7a	189 $\pm$ 18.6a	148 $\pm$ 16.5a	1.1	0.3558
g CH ₄ kg LW ⁻¹	0.54 $\pm$ 0.04a	0.63 $\pm$ 0.06a	0.58 $\pm$ 0.05a	0.3	0.7697
g CH ₄ kg ADG ⁻¹	279 $\pm$ 31.1a	259 $\pm$ 25.0ab	183 $\pm$ 26.4b	4.3	0.0259
kg CH ₄ ha ⁻¹ day ⁻¹	0.71 $\pm$ 0.080a	0.53 $\pm$ 0.058b	0.66 $\pm$ 0.078ab	3.5	0.0558

F and P = F-ratio and probability, respectively, for the year effect; SH = sward height; HM = herbage mass; HAR = herbage accumulation rate; HA = herbage allowance; SR = stocking rate; ADG = average daily gain; Gha = gain per hectare; LW = live weight; LP = leaf proportion; CH₄, enteric methane. Within rows, means (n = 12) followed by the same letter are not significantly different according to LSD test (P < 0.05).

diverse combination of plant species and parts, CH₄ production could be reduced more by selective feeding on forage with a better quality in grasslands than in communities containing only a few grass species. Emphasizing grazing on plant communities with higher species richness is a potential strategy for the mitigation of greenhouse gas emissions (Hammond et al., 2014).

The inclusion of trees in integrated crop-livestock systems resulted in a reduction in pasture growth after 6 years of tree growth. This reduced pasture growth led to a decrease in animal production but was not associated with increased CH₄ emissions per animal in either grazing year. This result agrees with Pinares-Patiño et al. (2006), who did not find differences in absolute CH₄ emission (g day⁻¹) despite significant differences in HM and the nutritive value of a semi-natural grassland managed at low or high SR.

However, differences were observed in terms of CH₄ emission per ADG, which suggest that CH₄ production is reduced in heifers experiencing large live-weight gains (Patra, 2012). The methane production of heifers grazing in the CLT N90 treatment in 2012/2013 (i.e., the treatment and years with lower ADG) was 1.6-fold higher than heifers grazing in the CL N180 or in 2014 (Tables 4 and 5). This increase in CH₄ production could reflect a decrease in production efficiency in the CLT system. In terms of CO₂eqv, the values obtained in this study were 4.2 and 6.0 kg CO₂eqv ADG⁻¹ for the CL and CLT systems, respectively. Phetteplace et al. (2001) reported values of 8.4 and 6.4 kg CO₂eqv ADG⁻¹ for beef cows and grazing cattle after weaning, respectively. Therefore, cattle in this study were more efficient and produced more per unit CO₂eqv emitted.

Our results indicate lower CH₄ emissions per unit area for the CLT compared to the CL systems and lower emissions in 2013 than in 2012 and 2014. The latter result was due to variations in SR (Table 5). Pasture shading by trees led to an average reduction in total annual herbage accumulation of approximately 2.6 Mg DM ha⁻¹ and an average reduction in winter carrying capacity of 0.5 stock units ha⁻¹, which led to a lower animal stocking rate and lower CH₄ emissions. The higher CH₄ emission per unit area in the CL system (+54%) was compensated by a higher addition of residues (> HM), which in turn influences soil properties and the resulting greenhouse gas budget.

4.3. Methane mitigation in crop-livestock-tree systems

As noted above, lower productivity in the CLT systems indicates the need for more intensive silvicultural interventions. For instance, a tree species (e.g., *G. robusta*) could be thinned earlier (at 6 years of age) to improve the penetration of solar irradiance to the alley crop by reducing the shading level to < 39% and tree density to 79 trees ha⁻¹. The remaining eucalyptus (*E. dunnii*) trees would then produce better-quality wood. Therefore, to estimate C sequestration in the woody biomass of trees, we will consider only the eucalyptus trees because the C stored by these trees could remain in the system longer.

Annual CH₄ emissions from enteric fermentation, using values per unit ground area, were 326 kg Ceqv ha⁻¹ year⁻¹ for the CLT system. This value was obtained assuming a grazing period of approximately 100 days per year on areas with integrated crop-livestock systems. The average diameter at breast height and average height of eucalyptus trees in October 2014 at 8 years of age was 43 cm and 22 m, respectively. From these values, we calculated the volume of each tree as 1.6 m³. This volume can be transformed to C by multiplying it per woody basic density (e.g., 500 kg m⁻³ for *E. dunnii* according Paludyszyn Filho and Santos, 2005) and by 0.42 because it is assumed that 42% of the woody biomass is composed of C (Oliveira, 2011). Therefore, each 8-year-old eucalyptus tree has the potential to sequester an average of 336 kg of C. In other words, one tree has the potential to mitigate 100 days of CH₄ emissions from animals (with 2.1 stock units ha⁻¹ on average). With 79 eucalyptus trees per hectare in 2014, we sequestered 26.5 Mg of C per hectare. Because C sequestration in the woody biomass of trees is very large, the amount of C sequestered

per unit area is substantially higher in the CLT systems compared to conventional livestock systems. In that case, it should be possible to neutralize the CH₄ emissions from cattle kept in these systems with a carbon surplus of ~26.2 Mg C ha⁻¹ or 96.1 Mg CO₂eqv. ha⁻¹. These systems could be designed to protect and enhance C sequestration, particularly in sensitive landscapes such as degraded sites with water pollution potential (Udawatta and Jose, 2012).

The integrated crop-livestock systems described here can be used for finishing animals. However, summer pastures in association with winter species can supply forage throughout the year. Therefore, additional research efforts are required to enhance our understanding of CH₄ emissions per unit of animal product from integrated systems.

5. Conclusions

The 6–8-year-old eucalyptus and silver oak trees, with a planting density of 159 trees ha⁻¹, reduced beef heifer gains on a black oat + ryegrass mixture by 32% compared to pastures without trees. An increase in N level did not compensate for these losses. Therefore, silvicultural interventions need to be intensified to reduce the shading level to below 39% and avoid losses to animal intake and soil cover associated with sub-optimal herbage mass. Shading also led to higher greenhouse gas emissions in the CLT systems, as daily CH₄ production per kg of gain is greater in heifers experiencing lower live-weight gains. Improving the penetration of solar irradiance to the alley crop allows for higher pasture productivity but increases CH₄ emissions per area due to the consequent increase in the stocking rate. However, our estimates clearly show that the high C sequestration potential of woody biomass more than compensates for increased emissions per unit area in the CLT systems, which makes integrated crop-livestock systems with trees a promising CO₂ mitigation strategy in subtropical regions.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.agee.2017.11.009>.

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